



## 1. Description

### 1.1. Project

|                 |                    |
|-----------------|--------------------|
| Project Name    | g4-oscilloscope-b  |
| Board Name      | NUCLEO-G474RE      |
| Generated with: | STM32CubeMX 6.17.0 |
| Date            | 06/30/2026         |

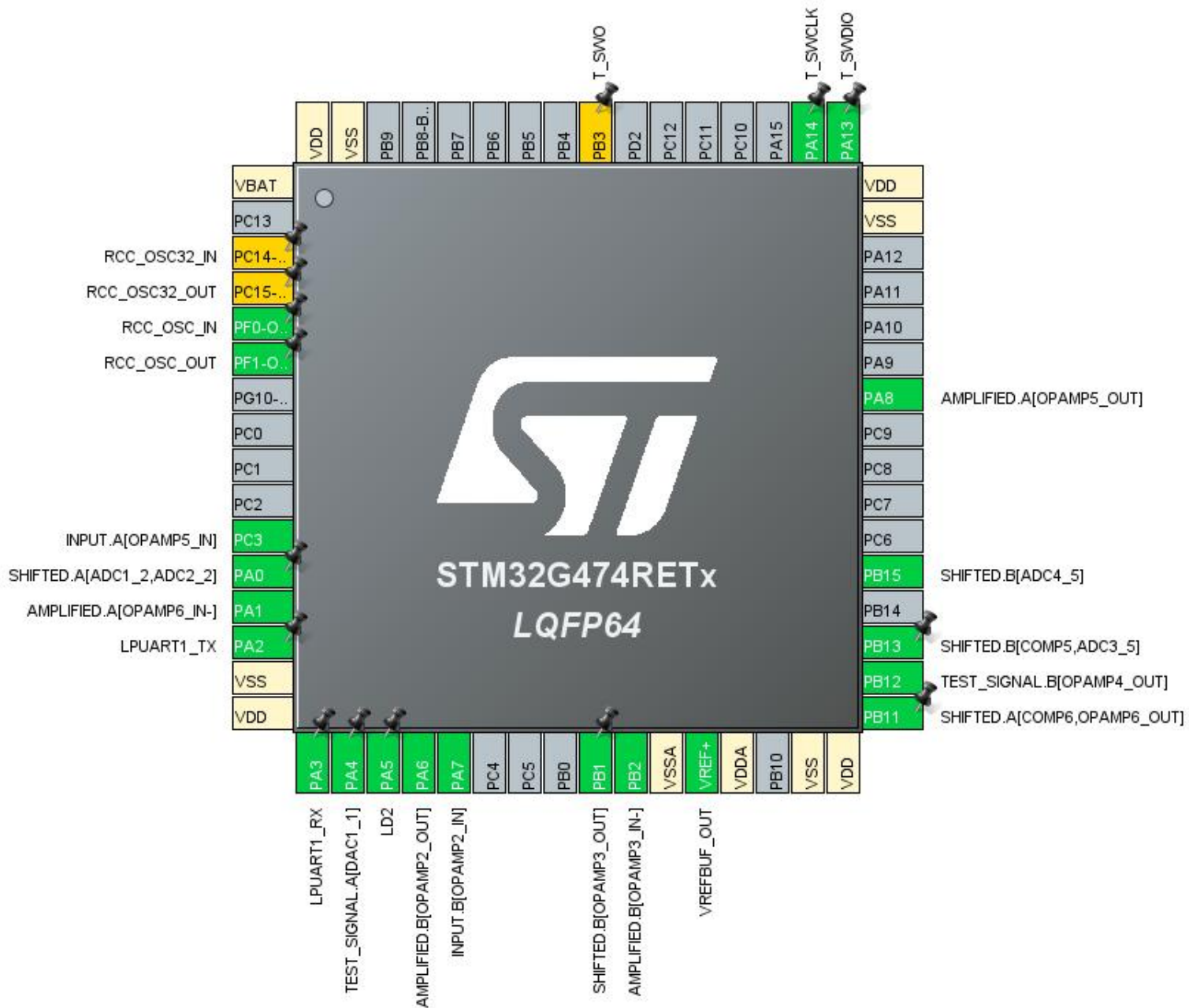
### 1.2. MCU

|                |               |
|----------------|---------------|
| MCU Series     | STM32G4       |
| MCU Line       | STM32G4x4     |
| MCU name       | STM32G474RETx |
| MCU Package    | LQFP64        |
| MCU Pin number | 64            |

### 1.3. Core(s) information

|         |               |
|---------|---------------|
| Core(s) | ARM Cortex-M4 |
|---------|---------------|

## 2. Pinout Configuration



### 3. Pins Configuration

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s)  | Label                       |
|----------------------|---------------------------------------|----------|---------------------------|-----------------------------|
| 1                    | VBAT                                  | Power    |                           |                             |
| 3                    | PC14-OSC32_IN *                       | I/O      | RCC_OSC32_IN              | RCC_OSC32_IN                |
| 4                    | PC15-OSC32_OUT *                      | I/O      | RCC_OSC32_OUT             | RCC_OSC32_OUT               |
| 5                    | PF0-OSC_IN                            | I/O      | RCC_OSC_IN                | RCC_OSC_IN                  |
| 6                    | PF1-OSC_OUT                           | I/O      | RCC_OSC_OUT               | RCC_OSC_OUT                 |
| 11                   | PC3                                   | I/O      | OPAMP5_VINP               | INPUT.A[OPAMP5_IN]          |
| 12                   | PA0                                   | I/O      | ADC1_IN1, ADC2_IN1        | SHIFTED.A[ADC1_2,ADC2_2]    |
| 13                   | PA1                                   | I/O      | OPAMP6_VINM0              | AMPLIFIED.A[OPAMP6_IN-]     |
| 14                   | PA2                                   | I/O      | LPUART1_TX                |                             |
| 15                   | VSS                                   | Power    |                           |                             |
| 16                   | VDD                                   | Power    |                           |                             |
| 17                   | PA3                                   | I/O      | LPUART1_RX                |                             |
| 18                   | PA4                                   | I/O      | DAC1_OUT1                 | TEST_SIGNAL.A[DAC1_1]       |
| 19                   | PA5 **                                | I/O      | GPIO_Output               | LD2                         |
| 20                   | PA6                                   | I/O      | OPAMP2_VOUT               | AMPLIFIED.B[OPAMP2_OUT]     |
| 21                   | PA7                                   | I/O      | OPAMP2_VINP               | INPUT.B[OPAMP2_IN]          |
| 25                   | PB1                                   | I/O      | OPAMP3_VOUT               | SHIFTED.B[OPAMP3_OUT]       |
| 26                   | PB2                                   | I/O      | OPAMP3_VINM0              | AMPLIFIED.B[OPAMP3_IN-]     |
| 27                   | VSSA                                  | Power    |                           |                             |
| 28                   | VREF+                                 | MonoIO   | VREFBUF_OUT               |                             |
| 29                   | VDDA                                  | Power    |                           |                             |
| 31                   | VSS                                   | Power    |                           |                             |
| 32                   | VDD                                   | Power    |                           |                             |
| 33                   | PB11                                  | I/O      | OPAMP6_VOUT,<br>COMP6_INP | SHIFTED.A[COMP6,OPAMP6_OUT] |
| 34                   | PB12                                  | I/O      | OPAMP4_VOUT               | TEST_SIGNAL.B[OPAMP4_OUT]   |
| 35                   | PB13                                  | I/O      | ADC3_IN5, COMP5_INP       | SHIFTED.B[COMP5,ADC3_5]     |
| 37                   | PB15                                  | I/O      | ADC4_IN5                  | SHIFTED.B[ADC4_5]           |
| 42                   | PA8                                   | I/O      | OPAMP5_VOUT               | AMPLIFIED.A[OPAMP5_OUT]     |
| 47                   | VSS                                   | Power    |                           |                             |

| Pin Number<br>LQFP64 | Pin Name<br>(function after<br>reset) | Pin Type | Alternate<br>Function(s) | Label   |
|----------------------|---------------------------------------|----------|--------------------------|---------|
| 48                   | VDD                                   | Power    |                          |         |
| 49                   | PA13                                  | I/O      | SYS_JTMS-SWDIO           | T_SWDIO |
| 50                   | PA14                                  | I/O      | SYS_JTCK-SWCLK           | T_SWCLK |
| 56                   | PB3 *                                 | I/O      | SYS_JTDO-SWO             | T_SWO   |
| 63                   | VSS                                   | Power    |                          |         |
| 64                   | VDD                                   | Power    |                          |         |

\*\* The pin is affected with an I/O function

\* The pin is affected with a peripheral function but no peripheral mode is activated



## 1. Power Consumption Calculator report

### 1.1. Microcontroller Selection

|           |               |
|-----------|---------------|
| Series    | STM32G4       |
| Line      | STM32G4x4     |
| MCU       | STM32G474RETx |
| Datasheet | DS12288 Rev0  |

### 1.2. Parameter Selection

|             |     |
|-------------|-----|
| Temperature | 25  |
| Vdd         | 3.0 |

### 1.3. Battery Selection

|                   |                 |
|-------------------|-----------------|
| Battery           | Li-SOCL2(A3400) |
| Capacity          | 3400.0 mAh      |
| Self Discharge    | 0.08 %/month    |
| Nominal Voltage   | 3.6 V           |
| Max Cont Current  | 100.0 mA        |
| Max Pulse Current | 200.0 mA        |
| Cells in series   | 1               |
| Cells in parallel | 1               |

#### 1.4. Sequence

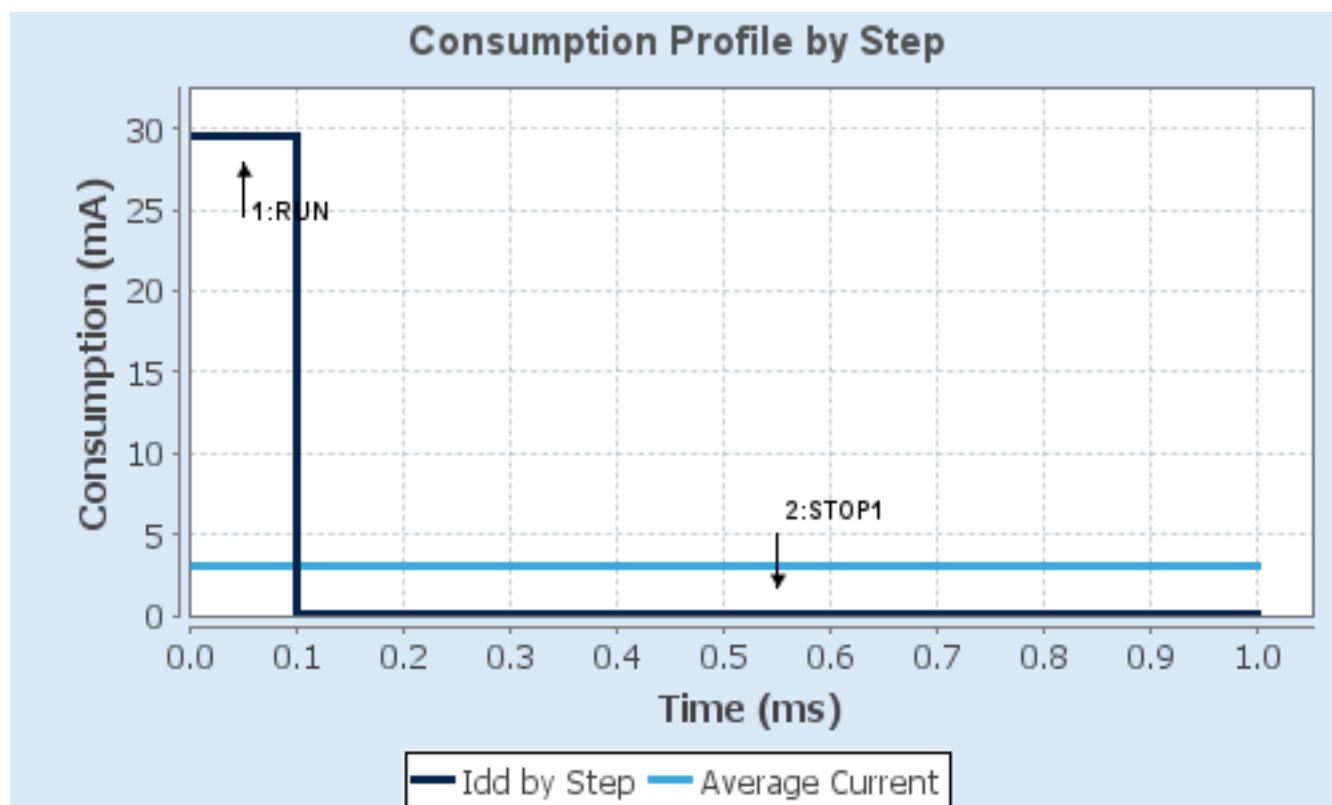
|                               |                    |                |
|-------------------------------|--------------------|----------------|
| <b>Step</b>                   | Step1              | Step2          |
| <b>Mode</b>                   | RUN                | STOP1          |
| <b>Vdd</b>                    | 3.0                | 3.0            |
| <b>Voltage Source</b>         | Battery            | Battery        |
| <b>Range</b>                  | Range1-Boost       | NoRange        |
| <b>Fetch Type</b>             | FLASH/DualBank/ART | NA             |
| <b>CPU Frequency</b>          | 170 MHz            | 0 Hz           |
| <b>Clock Configuration</b>    | HSE BYP PLL        | ALL CLOCKS OFF |
| <b>Clock Source Frequency</b> | 4 MHz              | 0 Hz           |
| <b>Peripherals</b>            |                    |                |
| <b>Additional Cons.</b>       | 0 mA               | 0 mA           |
| <b>Average Current</b>        | 29.5 mA            | 80.5 $\mu$ A   |
| <b>Duration</b>               | 0.1 ms             | 0.9 ms         |
| <b>DMIPS</b>                  | 213.0              | 0.0            |
| <b>Ta Max</b>                 | 124.25             | 129.98         |
| <b>Category</b>               | In DS Table        | In DS Table    |

#### 1.5. Results

|               |                              |                 |             |
|---------------|------------------------------|-----------------|-------------|
| Sequence Time | 1 ms                         | Average Current | 3.02 mA     |
| Battery Life  | 1 month, 16 days,<br>9 hours | Average DMIPS   | 212.5 DMIPS |

#### 1.6. Chart





## 2. Software Project

### 2.1. Project Settings

| Name                              | Value                         |
|-----------------------------------|-------------------------------|
| Project Name                      | g4-oscilloscope-b             |
| Project Folder                    | D:\projects\g4-oscilloscope-b |
| Toolchain / IDE                   | CMake                         |
| Firmware Package Name and Version | STM32Cube FW_G4 V1.6.3        |
| Application Structure             | Advanced                      |
| Generate Under Root               | No                            |
| Do not generate the main()        | No                            |
| Minimum Heap Size                 | 0x200                         |
| Minimum Stack Size                | 0x400                         |

### 2.2. Code Generation Settings

| Name                                                            | Value                                 |
|-----------------------------------------------------------------|---------------------------------------|
| STM32Cube MCU packages and embedded software                    | Copy only the necessary library files |
| Generate peripheral initialization as a pair of '.c/.h' files   | Yes                                   |
| Backup previously generated files when re-generating            | No                                    |
| Keep User Code when re-generating                               | Yes                                   |
| Delete previously generated files when not re-generated         | Yes                                   |
| Set all free pins as analog (to optimize the power consumption) | No                                    |
| Enable Full Assert                                              | No                                    |

### 2.3. Advanced Settings - Generated Function Calls

| Rank | Function Name        | Peripheral Instance Name |
|------|----------------------|--------------------------|
| 1    | MX_GPIO_Init         | GPIO                     |
| 2    | MX_DMA_Init          | DMA                      |
| 3    | SystemClock_Config   | RCC                      |
| 4    | MX_ADC3_Init         | ADC3                     |
| 5    | MX_OPAMP3_Init       | OPAMP3                   |
| 6    | MX_DAC1_Init         | DAC1                     |
| 7    | MX_ADC4_Init         | ADC4                     |
| 8    | MX_LPUART1_UART_Init | LPUART1                  |
| 9    | MX_TIM1_Init         | TIM1                     |
| 10   | MX_TIM15_Init        | TIM15                    |
| 11   | MX_TIM2_Init         | TIM2                     |

| Rank | Function Name  | Peripheral Instance Name |
|------|----------------|--------------------------|
| 12   | MX_DAC4_Init   | DAC4                     |
| 13   | MX_OPAMP5_Init | OPAMP5                   |
| 14   | MX_DAC3_Init   | DAC3                     |
| 15   | MX_ADC1_Init   | ADC1                     |
| 16   | MX_ADC2_Init   | ADC2                     |
| 17   | MX_COMP5_Init  | COMP5                    |
| 18   | MX_COMP6_Init  | COMP6                    |
| 19   | MX_OPAMP2_Init | OPAMP2                   |
| 20   | MX_OPAMP6_Init | OPAMP6                   |
| 21   | MX_OPAMP4_Init | OPAMP4                   |

## 3. Peripherals and Middlewares Configuration

### 3.1. ADC1

#### IN1: IN1 Single-ended

#### mode: Vrefint Channel

##### 3.1.1. Parameter Settings:

##### **ADCs\_Common\_Settings:**

Mode

**Dual Interleaved mode + injected simultaneous mode \***

DMA Access Mode

DMA access mode disabled

Delay between 2 sampling phases

**6 Cycles \***

##### **ADC\_Settings:**

Clock Prescaler

**Asynchronous clock mode divided by 1 \***

Resolution

ADC 12-bit resolution

Data Alignment

Right alignment

Gain Compensation

0

Scan Conversion Mode

Disabled

End Of Conversion Selection

End of single conversion

Low Power Auto Wait

Disabled

Continuous Conversion Mode

Disabled

Discontinuous Conversion Mode

Disabled

DMA Continuous Requests

**Enabled \***

Overrun behaviour

**Overrun data overwritten \***

##### **ADC\_Regular\_ConversionMode:**

Enable Regular Conversions

Enable

Enable Regular Oversampling

Disable

Number Of Conversion

1

External Trigger Conversion Source

**Timer 2 Trigger Out event \***

External Trigger Conversion Edge

Trigger detection on the rising edge

Rank

1

Channel

Channel 1

Sampling Time

2.5 Cycles

Offset Number

No offset

##### **ADC\_Injected\_ConversionMode:**

Enable Injected Conversions

**Enable \***

Number Of Conversions

**1 \***

External Trigger Source

Injected Conversion launched by software

External Trigger Conversion Edge

None

Enable Injected Oversampling

Disable

Injected Conversion Mode

None

|                |                          |
|----------------|--------------------------|
| Injected Queue | Injected Queue Disable   |
| <u>Rank</u>    | 1                        |
| Channel        | <b>Channel Vrefint *</b> |
| Sampling Time  | <b>640.5 Cycles *</b>    |
| Offset Number  | No offset                |

#### **Analog Watchdog 1:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog1 Mode | false |
|------------------------------|-------|

#### **Analog Watchdog 2:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog2 Mode | false |
|------------------------------|-------|

#### **Analog Watchdog 3:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog3 Mode | false |
|------------------------------|-------|

## **3.2. ADC2**

### **IN1: IN1 Single-ended**

#### 3.2.1. Parameter Settings:

##### **ADCs\_Common\_Settings:**

|                                 |                                                             |
|---------------------------------|-------------------------------------------------------------|
| Mode                            | <b>Dual Interleaved mode + injected simultaneous mode *</b> |
| DMA Access Mode                 | DMA access mode disabled                                    |
| Delay between 2 sampling phases | <b>6 Cycles *</b>                                           |

##### **ADC\_Settings:**

|                               |                                               |
|-------------------------------|-----------------------------------------------|
| Clock Prescaler               | <b>Asynchronous clock mode divided by 1 *</b> |
| Resolution                    | ADC 12-bit resolution                         |
| Data Alignment                | Right alignment                               |
| Gain Compensation             | 0                                             |
| Scan Conversion Mode          | Disabled                                      |
| End Of Conversion Selection   | End of single conversion                      |
| Low Power Auto Wait           | Disabled                                      |
| Continuous Conversion Mode    | Disabled                                      |
| Discontinuous Conversion Mode | Disabled                                      |
| DMA Continuous Requests       | Disabled                                      |
| Overrun behaviour             | Overrun data preserved                        |

##### **ADC\_Regular\_ConversionMode:**

|                             |            |
|-----------------------------|------------|
| Enable Regular Conversions  | Enable     |
| Enable Regular Oversampling | Disable    |
| Number Of Conversion        | 1          |
| <u>Rank</u>                 | 1          |
| Channel                     | Channel 1  |
| Sampling Time               | 2.5 Cycles |

Offset Number No offset

#### **ADC\_Injected\_ConversionMode:**

Enable Injected Conversions Disable

#### **Analog Watchdog 1:**

Enable Analog WatchDog1 Mode false

#### **Analog Watchdog 2:**

Enable Analog WatchDog2 Mode false

#### **Analog Watchdog 3:**

Enable Analog WatchDog3 Mode false

### **3.3. ADC3**

#### **mode: IN5**

##### 3.3.1. Parameter Settings:

#### **ADCs\_Common\_Settings:**

Mode

DMA Access Mode

Delay between 2 sampling phases

**Dual interleaved mode only \***

DMA access mode enabled

**6 Cycles \***

#### **ADC\_Settings:**

Clock Prescaler

Resolution

Data Alignment

Gain Compensation

Scan Conversion Mode

End Of Conversion Selection

Low Power Auto Wait

Continuous Conversion Mode

Discontinuous Conversion Mode

DMA Continuous Requests

Overrun behaviour

**Asynchronous clock mode divided by 1 \***

ADC 12-bit resolution

Right alignment

0

Disabled

End of single conversion

Disabled

Disabled

Disabled

**Enabled \***

**Overrun data overwritten \***

#### **ADC\_Regular\_ConversionMode:**

Enable Regular Conversions

Enable Regular Oversampling

Number Of Conversion

External Trigger Conversion Source

External Trigger Conversion Edge

Rank

Channel

Enable

Disable

1

**Timer 2 Trigger Out event \***

Trigger detection on the rising edge

1

Channel 5

|               |            |
|---------------|------------|
| Sampling Time | 2.5 Cycles |
| Offset Number | No offset  |

#### **ADC\_Injected\_ConversionMode:**

|                             |         |
|-----------------------------|---------|
| Enable Injected Conversions | Disable |
|-----------------------------|---------|

#### **Analog Watchdog 1:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog1 Mode | false |
|------------------------------|-------|

#### **Analog Watchdog 2:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog2 Mode | false |
|------------------------------|-------|

#### **Analog Watchdog 3:**

|                              |       |
|------------------------------|-------|
| Enable Analog WatchDog3 Mode | false |
|------------------------------|-------|

### **3.4. ADC4**

#### **mode: IN5**

##### 3.4.1. Parameter Settings:

#### **ADCs\_Common\_Settings:**

|                                 |                                     |
|---------------------------------|-------------------------------------|
| Mode                            | <b>Dual interleaved mode only *</b> |
| DMA Access Mode                 | DMA access mode enabled             |
| Delay between 2 sampling phases | <b>6 Cycles *</b>                   |

#### **ADC\_Settings:**

|                               |                                               |
|-------------------------------|-----------------------------------------------|
| Clock Prescaler               | <b>Asynchronous clock mode divided by 1 *</b> |
| Resolution                    | ADC 12-bit resolution                         |
| Data Alignment                | Right alignment                               |
| Gain Compensation             | 0                                             |
| Scan Conversion Mode          | Disabled                                      |
| End Of Conversion Selection   | End of single conversion                      |
| Low Power Auto Wait           | Disabled                                      |
| Continuous Conversion Mode    | Disabled                                      |
| Discontinuous Conversion Mode | Disabled                                      |
| DMA Continuous Requests       | Disabled                                      |
| Overrun behaviour             | <b>Overrun data overwritten *</b>             |

#### **ADC\_Regular\_ConversionMode:**

|                             |            |
|-----------------------------|------------|
| Enable Regular Conversions  | Enable     |
| Enable Regular Oversampling | Disable    |
| Number Of Conversion        | 1          |
| <u>Rank</u>                 | 1          |
| Channel                     | Channel 5  |
| Sampling Time               | 2.5 Cycles |
| Offset Number               | No offset  |

**ADC\_Injected\_ConversionMode:**

Enable Injected Conversions                      Disable

**Analog Watchdog 1:**

Enable Analog WatchDog1 Mode                      false

**Analog Watchdog 2:**

Enable Analog WatchDog2 Mode                      false

**Analog Watchdog 3:**

Enable Analog WatchDog3 Mode                      false

### 3.5. COMP5

**mode: Input [ + ]**

**Input [-]: DAC1 OUT2**

3.5.1. Parameter Settings:

**Basic Parameters:**

Trigger Mode

**Rising/Falling Edge Interrupt \***

Hysteresis Level

**Level 10mV \***

**Output Configuration:**

Blanking Source

None

Output Polarity

COMP output on GPIO isn't inverted

### 3.6. COMP6

**mode: Input [ + ]**

**Input [-]: DAC4 OUT2**

3.6.1. Parameter Settings:

**Basic Parameters:**

Trigger Mode

**Rising/Falling Edge Interrupt \***

Hysteresis Level

**Level 10mV \***

**Output Configuration:**

Blanking Source

None

Output Polarity

COMP output on GPIO isn't inverted



### 3.7. DAC1

**OUT1 mode: Connected to external pin only**

**OUT2 mode: OUT2 Connected to on chip-peripherals only**

#### 3.7.1. Parameter Settings:

##### **DAC Out1 Settings:**

|                      |                                     |
|----------------------|-------------------------------------|
| Mode selected        | Normal Mode                         |
| Output Buffer        | Enable                              |
| DAC High Frequency   | Mode Automatic                      |
| DMA Double Data      | <b>Enable *</b>                     |
| Signed Format        | Disable                             |
| Trigger              | <b>Timer 15 Trigger Out event *</b> |
| Trigger2             | None                                |
| Wave generation mode | Disabled                            |
| User Trimming        | Factory trimming                    |

##### **DAC Out2 Settings:**

|                    |                  |
|--------------------|------------------|
| Mode selected      | Normal Mode      |
| Output Buffer      | Disable          |
| DAC High Frequency | Mode Automatic   |
| DMA Double Data    | Disable          |
| Signed Format      | Disable          |
| Trigger            | None             |
| Trigger2           | None             |
| User Trimming      | Factory trimming |

### 3.8. DAC3

**mode: OUT1 mode**

**mode: OUT2 mode**

#### 3.8.1. Parameter Settings:

##### **DAC Out1 Settings:**

|                    |                  |
|--------------------|------------------|
| Mode selected      | Normal Mode      |
| Output Buffer      | Disable          |
| DAC High Frequency | Mode Automatic   |
| DMA Double Data    | Disable          |
| Signed Format      | Disable          |
| Trigger            | None             |
| Trigger2           | None             |
| User Trimming      | Factory trimming |

#### DAC Out2 Settings:

|                    |                  |
|--------------------|------------------|
| Mode selected      | Normal Mode      |
| Output Buffer      | Disable          |
| DAC High Frequency | Mode Automatic   |
| DMA Double Data    | Disable          |
| Signed Format      | Disable          |
| Trigger            | None             |
| Trigger2           | None             |
| User Trimming      | Factory trimming |

### 3.9. DAC4

**mode: OUT1 mode**

**mode: OUT2 mode**

#### 3.9.1. Parameter Settings:

#### DAC Out1 Settings:

|                    |                |
|--------------------|----------------|
| Mode selected      | Normal Mode    |
| Output Buffer      | Disable        |
| DAC High Frequency | Mode Automatic |
| DMA Double Data    | Disable        |
| Signed Format      | Disable        |

Trigger

**Timer 15 Trigger Out event \***

Trigger2

None

Wave generation mode

**Triangle wave generation \***

Maximum Triangle Amplitude

**2047 \***

User Trimming

Factory trimming

#### DAC Out2 Settings:

|                    |                  |
|--------------------|------------------|
| Mode selected      | Normal Mode      |
| Output Buffer      | Disable          |
| DAC High Frequency | Mode Automatic   |
| DMA Double Data    | Disable          |
| Signed Format      | Disable          |
| Trigger            | None             |
| Trigger2           | None             |
| User Trimming      | Factory trimming |

### 3.10. LPUART1

## Mode: Asynchronous

### 3.10.1. Parameter Settings:

#### Basic Parameters:

|             |                           |
|-------------|---------------------------|
| Baud Rate   | <b>460800 *</b>           |
| Word Length | 8 Bits (including Parity) |
| Parity      | None                      |
| Stop Bits   | 1                         |

#### Advanced Parameters:

|                  |                                  |
|------------------|----------------------------------|
| Data Direction   | Receive and Transmit             |
| Single Sample    | Disable                          |
| ClockPrescaler   | 1                                |
| Fifo Mode        | <b>FIFO mode enable *</b>        |
| Txfifo Threshold | <b>half full configuration *</b> |
| Rxfifo Threshold | 1 eighth full configuration      |

#### Advanced Features:

|                               |         |
|-------------------------------|---------|
| TX Pin Active Level Inversion | Disable |
| RX Pin Active Level Inversion | Disable |
| Data Inversion                | Disable |
| TX and RX pins Swapping       | Disable |
| Overrun                       | Enable  |
| DMA on RX Error               | Enable  |
| MSB First                     | Disable |

## 3.11. OPAMP2

### Mode: Follower

### 3.11.1. Parameter Settings:

#### Basic Parameters:

|               |                     |
|---------------|---------------------|
| Power Mode    | <b>High Speed *</b> |
| User Trimming | Disable             |

## 3.12. OPAMP3

### Mode: PGA Connected-INVERTINGINPUT\_IO0\_BIAS-DAC3\_OUT2-INP

### 3.12.1. Parameter Settings:

**Basic Parameters:**

|               |                     |
|---------------|---------------------|
| Power Mode    | <b>High Speed *</b> |
| PGA Gain      | <b>16 or -15 *</b>  |
| User Trimming | Disable             |

**3.13. OPAMP4**

**Mode: Follower-DAC4\_OUT1-INP**

3.13.1. Parameter Settings:

**Basic Parameters:**

|               |                     |
|---------------|---------------------|
| Power Mode    | <b>High Speed *</b> |
| User Trimming | Disable             |

**3.14. OPAMP5**

**Mode: Follower**

3.14.1. Parameter Settings:

**Basic Parameters:**

|               |                     |
|---------------|---------------------|
| Power Mode    | <b>High Speed *</b> |
| User Trimming | Disable             |

**3.15. OPAMP6**

**Mode: PGA Connected-INVERTINGINPUT\_IO0\_BIAS-DAC3\_OUT1-INP**

3.15.1. Parameter Settings:

**Basic Parameters:**

|               |                     |
|---------------|---------------------|
| Power Mode    | <b>High Speed *</b> |
| PGA Gain      | 2 or -1             |
| User Trimming | Disable             |

**3.16. RCC**

## High Speed Clock (HSE): Crystal/Ceramic Resonator

### 3.16.1. Parameter Settings:

#### System Parameters:

|                   |                    |
|-------------------|--------------------|
| VDD voltage (V)   | 3.3                |
| Instruction Cache | Enabled            |
| Prefetch Buffer   | Disabled           |
| Data Cache        | Enabled            |
| Flash Latency(WS) | 4 WS (5 CPU cycle) |

#### RCC Parameters:

|                                |      |
|--------------------------------|------|
| HSI Calibration Value          | 64   |
| HSE Startup Timeout Value (ms) | 100  |
| LSE Startup Timeout Value (ms) | 5000 |

#### Power Parameters:

|                               |                                 |
|-------------------------------|---------------------------------|
| Power Regulator Voltage Scale | Power Regulator Voltage Scale 1 |
|-------------------------------|---------------------------------|

#### Peripherals Clock Configuration:

|                                              |      |
|----------------------------------------------|------|
| Generate the peripherals clock configuration | TRUE |
|----------------------------------------------|------|

## 3.17. SYS

### Debug: Serial Wire

### VREFBUF Mode: Internal voltage reference

### Timebase Source: TIM7

### mode: save power of non-active UCPD - deactive Dead Battery pull-up

### 3.17.1. Parameter Settings:

#### Voltage\_Reference\_Buffer\_Settings:

|                                  |                         |
|----------------------------------|-------------------------|
| Trimming Mode                    | Factory Trimming        |
| Internal Voltage reference scale | SCALE 0: around 2.048 V |

## 3.18. TIM1

### Slave Mode: External Clock Mode 1

### Trigger Source: ITR1

### Channel1: PWM Generation No Output

### mode: One Pulse Mode

### 3.18.1. Parameter Settings:

### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | 0           |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 16 bits value ) | 65535       |
| Internal Clock Division (CKD)                         | No Division |
| Repetition Counter (RCR - 16 bits value)              | 0           |
| auto-reload preload                                   | Disable     |
| Slave Mode Controller                                 | ETR mode 1  |

### Trigger Output (TRGO) Parameters:

|                               |                                            |
|-------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)   | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO  | <b>Output Compare (OC1REF) *</b>           |
| Trigger Event Selection TRGO2 | Reset (UG bit from TIMx_EGR)               |

### Break And Dead Time management - BRK Configuration:

|                           |         |
|---------------------------|---------|
| BRK State                 | Disable |
| BRK Polarity              | High    |
| BRK Filter (4 bits value) | 0       |
| BRK Sources Configuration |         |
| - Digital Input           | Disable |
| - COMP1                   | Disable |
| - COMP2                   | Disable |
| - COMP3                   | Disable |
| - COMP4                   | Disable |
| - COMP5                   | Disable |
| - COMP6                   | Disable |
| - COMP7                   | Disable |

### Break And Dead Time management - BRK2 Configuration:

|                            |         |
|----------------------------|---------|
| BRK2 State                 | Disable |
| BRK2 Polarity              | High    |
| BRK2 Filter (4 bits value) | 0       |
| BRK2 Sources Configuration |         |
| - Digital Input            | Disable |
| - COMP1                    | Disable |
| - COMP2                    | Disable |
| - COMP3                    | Disable |
| - COMP4                    | Disable |
| - COMP5                    | Disable |
| - COMP6                    | Disable |
| - COMP7                    | Disable |

### Break And Dead Time management - Output Configuration:

|                        |         |
|------------------------|---------|
| Automatic Output State | Disable |
|------------------------|---------|

|                                          |         |
|------------------------------------------|---------|
| Off State Selection for Run Mode (OSSR)  | Disable |
| Off State Selection for Idle Mode (OSSI) | Disable |
| Lock Configuration                       | Off     |

#### Clear Input:

|                    |         |
|--------------------|---------|
| Clear Input Source | Disable |
|--------------------|---------|

#### PWM Generation Channel 1:

|                        |              |
|------------------------|--------------|
| Mode                   | PWM mode 1   |
| Pulse (16 bits value)  | <b>99 *</b>  |
| Output compare preload | Enable       |
| Fast Mode              | Disable      |
| CH Polarity            | High         |
| CH Idle State          | <b>Set *</b> |

### 3.19. TIM2

**Slave Mode: Gated Mode**

**Trigger Source: ITR0**

**Clock Source : Internal Clock**

#### 3.19.1. Parameter Settings:

##### Counter Settings:

|                                                       |             |
|-------------------------------------------------------|-------------|
| Prescaler (PSC - 16 bits value)                       | <b>39 *</b> |
| Counter Mode                                          | Up          |
| Dithering                                             | Disable     |
| Counter Period (AutoReload Register - 32 bits value ) | <b>19 *</b> |
| Internal Clock Division (CKD)                         | No Division |
| auto-reload preload                                   | Disable     |
| Slave Mode Controller                                 | Gated Mode  |

##### Trigger Output (TRGO) Parameters:

|                              |                                            |
|------------------------------|--------------------------------------------|
| Master/Slave Mode (MSM bit)  | Disable (Trigger input effect not delayed) |
| Trigger Event Selection TRGO | <b>Update Event *</b>                      |

### 3.20. TIM15

**mode: Clock Source**

#### 3.20.1. Parameter Settings:

##### Counter Settings:

|                                                       |                                            |
|-------------------------------------------------------|--------------------------------------------|
| Prescaler (PSC - 16 bits value)                       | <b>30 *</b>                                |
| Counter Mode                                          | Up                                         |
| Dithering                                             | Disable                                    |
| Counter Period (AutoReload Register - 16 bits value ) | <b>16 *</b>                                |
| Internal Clock Division (CKD)                         | No Division                                |
| Repetition Counter (RCR - 8 bits value)               | 0                                          |
| auto-reload preload                                   | Disable                                    |
| <b>Trigger Output (TRGO) Parameters:</b>              |                                            |
| Master/Slave Mode (MSM bit)                           | Disable (Trigger input effect not delayed) |
| Trigger Event Selection                               | <b>Update Event *</b>                      |

### 3.21. FREERTOS

#### Interface: CMSIS\_V2

##### 3.21.1. Config parameters:

##### API:

|              |          |
|--------------|----------|
| FreeRTOS API | CMSIS v2 |
|--------------|----------|

##### Versions:

|                    |        |
|--------------------|--------|
| FreeRTOS version   | 10.3.1 |
| CMSIS-RTOS version | 2.1.3  |

##### MPU/FPU:

|            |          |
|------------|----------|
| ENABLE_MPU | Disabled |
| ENABLE_FPU | Disabled |

##### Kernel settings:

|                           |                 |
|---------------------------|-----------------|
| USE_PREEMPTION            | Enabled         |
| CPU_CLOCK_HZ              | SystemCoreClock |
| TICK_RATE_HZ              | 1000            |
| MAX_PRIORITIES            | 56              |
| USE_SB_COMPLETED_CALLBACK | 0               |
| USE_MINI_LIST_ITEM        | 1               |
| MINIMAL_STACK_SIZE        | 128             |
| MAX_TASK_NAME_LEN         | 16              |
| USE_16_BIT_TICKS          | Disabled        |
| IDLE_SHOULD_YIELD         | Enabled         |
| USE_MUTEXES               | Enabled         |
| USE_RECURSIVE_MUTEXES     | Enabled         |
| USE_COUNTING_SEMAPHORES   | Enabled         |
| QUEUE_REGISTRY_SIZE       | 8               |
| USE_APPLICATION_TASK_TAG  | Disabled        |



|                                   |          |
|-----------------------------------|----------|
| ENABLE_BACKWARD_COMPATIBILITY     | Enabled  |
| USE_PORT_OPTIMISED_TASK_SELECTION | Disabled |
| USE_TICKLESS_IDLE                 | Disabled |
| USE_TASK_NOTIFICATIONS            | Enabled  |
| RECORD_STACK_HIGH_ADDRESS         | Disabled |

#### Memory management settings:

|                          |                  |
|--------------------------|------------------|
| Memory Allocation        | Dynamic / Static |
| TOTAL_HEAP_SIZE          | <b>30720 *</b>   |
| Memory Management scheme | heap_4           |

#### Hook function related definitions:

|                              |          |
|------------------------------|----------|
| USE_IDLE_HOOK                | Disabled |
| USE_TICK_HOOK                | Disabled |
| USE_MALLOC_FAILED_HOOK       | Disabled |
| USE_DAEMON_TASK_STARTUP_HOOK | Disabled |
| CHECK_FOR_STACK_OVERFLOW     | Disabled |

#### Run time and task stats gathering related definitions:

|                                |                  |
|--------------------------------|------------------|
| GENERATE_RUN_TIME_STATS        | <b>Enabled *</b> |
| USE_TRACE_FACILITY             | Enabled          |
| USE_STATS_FORMATTING_FUNCTIONS | Disabled         |

#### Co-routine related definitions:

|                           |          |
|---------------------------|----------|
| USE_CO_ROUTINES           | Disabled |
| MAX_CO_ROUTINE_PRIORITIES | 2        |

#### Software timer definitions:

|                        |         |
|------------------------|---------|
| USE_TIMERS             | Enabled |
| TIMER_TASK_PRIORITY    | 2       |
| TIMER_QUEUE_LENGTH     | 10      |
| TIMER_TASK_STACK_DEPTH | 256     |

#### Interrupt nesting behaviour configuration:

|                                        |    |
|----------------------------------------|----|
| LIBRARY_LOWEST_INTERRUPT_PRIORITY      | 15 |
| LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY | 5  |

#### Added with 10.2.1 support:

|                            |          |
|----------------------------|----------|
| MESSAGE_BUFFER_LENGTH_TYPE | size_t   |
| USE_POSIX_ERRNO            | Disabled |

#### CMSIS-RTOS V2 flags:

|                               |         |
|-------------------------------|---------|
| USE_OS2_THREAD_SUSPEND_RESUME | Enabled |
| USE_OS2_THREAD_ENUMERATE      | Enabled |
| USE_OS2_EVENTFLAGS_FROM_ISR   | Enabled |
| USE_OS2_THREAD_FLAGS          | Enabled |
| USE_OS2_TIMER                 | Enabled |
| USE_OS2_MUTEX                 | Enabled |

### 3.21.2. Include parameters:

#### **Include definitions:**

|                              |          |
|------------------------------|----------|
| vTaskPrioritySet             | Enabled  |
| uxTaskPriorityGet            | Enabled  |
| vTaskDelete                  | Enabled  |
| vTaskCleanUpResources        | Disabled |
| vTaskSuspend                 | Enabled  |
| vTaskDelayUntil              | Enabled  |
| vTaskDelay                   | Enabled  |
| xTaskGetSchedulerState       | Enabled  |
| xTaskResumeFromISR           | Enabled  |
| xQueueGetMutexHolder         | Enabled  |
| xSemaphoreGetMutexHolder     | Disabled |
| pcTaskGetTaskName            | Disabled |
| uxTaskGetStackHighWaterMark  | Enabled  |
| xTaskGetCurrentTaskHandle    | Enabled  |
| eTaskGetState                | Enabled  |
| xEventGroupSetBitFromISR     | Disabled |
| xTimerPendFunctionCall       | Enabled  |
| xTaskAbortDelay              | Disabled |
| xTaskGetHandle               | Disabled |
| uxTaskGetStackHighWaterMark2 | Disabled |

### 3.21.3. Advanced settings:

#### **Newlib settings (see parameter description first):**

|                      |          |
|----------------------|----------|
| USE_NEWLIB_REENTRANT | Disabled |
|----------------------|----------|

#### **Project settings (see parameter description first):**

|                       |         |
|-----------------------|---------|
| Use FW pack heap file | Enabled |
|-----------------------|---------|

**\* User modified value**

## 4. System Configuration

### 4.1. GPIO configuration

| IP      | Pin         | Signal         | GPIO mode                    | GPIO pull/up pull down      | Max Speed   | User Label                  |
|---------|-------------|----------------|------------------------------|-----------------------------|-------------|-----------------------------|
| ADC1    | PA0         | ADC1_IN1       | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.A[ADC1_2,ADC2_2]    |
| ADC2    | PA0         | ADC2_IN1       | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.A[ADC1_2,ADC2_2]    |
| ADC3    | PB13        | ADC3_IN5       | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.B[COMP5,ADC3_5]     |
| ADC4    | PB15        | ADC4_IN5       | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.B[ADC4_5]           |
| COMP5   | PB13        | COMP5_INP      | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.B[COMP5,ADC3_5]     |
| COMP6   | PB11        | COMP6_INP      | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.A[COMP6,OPAMP6_OUT] |
| DAC1    | PA4         | DAC1_OUT1      | Analog mode                  | No pull-up and no pull-down | n/a         | TEST_SIGNAL.A[DAC1_1]       |
| LPUART1 | PA2         | LPUART1_TX     | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |                             |
|         | PA3         | LPUART1_RX     | Alternate Function Push Pull | No pull-up and no pull-down | Very High * |                             |
| OPAMP2  | PA6         | OPAMP2_VOUT    | Analog mode                  | No pull-up and no pull-down | n/a         | AMPLIFIED.B[OPAMP2_OUT]     |
|         | PA7         | OPAMP2_VINP    | Analog mode                  | No pull-up and no pull-down | n/a         | INPUT.B[OPAMP2_IN]          |
| OPAMP3  | PB1         | OPAMP3_VOUT    | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.B[OPAMP3_OUT]       |
|         | PB2         | OPAMP3_VINM0   | Analog mode                  | No pull-up and no pull-down | n/a         | AMPLIFIED.B[OPAMP3_IN]      |
| OPAMP4  | PB12        | OPAMP4_VOUT    | Analog mode                  | No pull-up and no pull-down | n/a         | TEST_SIGNAL.B[OPAMP4_OUT]   |
| OPAMP5  | PC3         | OPAMP5_VINP    | Analog mode                  | No pull-up and no pull-down | n/a         | INPUT.A[OPAMP5_IN]          |
|         | PA8         | OPAMP5_VOUT    | Analog mode                  | No pull-up and no pull-down | n/a         | AMPLIFIED.A[OPAMP5_OUT]     |
| OPAMP6  | PA1         | OPAMP6_VINM0   | Analog mode                  | No pull-up and no pull-down | n/a         | AMPLIFIED.A[OPAMP6_IN]      |
|         | PB11        | OPAMP6_VOUT    | Analog mode                  | No pull-up and no pull-down | n/a         | SHIFTED.A[COMP6,OPAMP6_OUT] |
| RCC     | PF0-OSC_IN  | RCC_OSC_IN     | n/a                          | n/a                         | n/a         | RCC_OSC_IN                  |
|         | PF1-OSC_OUT | RCC_OSC_OUT    | n/a                          | n/a                         | n/a         | RCC_OSC_OUT                 |
| SYS     | VREF+       | VREFBUF_OUT    | n/a                          | n/a                         | n/a         |                             |
|         | PA13        | SYS_JTMS-SWDIO | n/a                          | n/a                         | n/a         | T_SWDIO                     |
|         | PA14        | SYS_JTCK-      | n/a                          | n/a                         | n/a         | T_SWCLK                     |

| IP                    | Pin            | Signal        | GPIO mode        | GPIO pull/up pull down      | Max Speed | User Label    |
|-----------------------|----------------|---------------|------------------|-----------------------------|-----------|---------------|
|                       |                | SWCLK         |                  |                             |           |               |
| Single Mapped Signals | PC14-OSC32_IN  | RCC_OSC32_IN  | n/a              | n/a                         | n/a       | RCC_OSC32_IN  |
|                       | PC15-OSC32_OUT | RCC_OSC32_OUT | n/a              | n/a                         | n/a       | RCC_OSC32_OUT |
|                       | PB3            | SYS_JTDO-SWO  | n/a              | n/a                         | n/a       | T_SWO         |
| GPIO                  | PA5            | GPIO_Output   | Output Push Pull | No pull-up and no pull-down | Low       | LD2           |

## 4.2. DMA configuration

| DMA request | Stream        | Direction            | Priority           |
|-------------|---------------|----------------------|--------------------|
| MEMTOMEM    | DMA1_Channel2 | Memory To Memory     | Low                |
| ADC3        | DMA1_Channel3 | Peripheral To Memory | Low                |
| LPUART1_TX  | DMA1_Channel4 | Memory To Peripheral | Low                |
| ADC1        | DMA1_Channel5 | Peripheral To Memory | Low                |
| MEMTOMEM    | DMA1_Channel6 | Memory To Memory     | Low                |
| DAC1_CH1    | DMA1_Channel1 | Memory To Peripheral | <b>Very High *</b> |

### MEMTOMEM: DMA1\_Channel2 DMA request Settings:

Mode: Normal  
 Src Memory Increment: **Enable \***  
 Dst Memory Increment: **Enable \***  
 Src Memory Data Width: **Word \***  
 Dst Memory Data Width: **Word \***

### ADC3: DMA1\_Channel3 DMA request Settings:

Mode: **Circular \***  
 Peripheral Increment: Disable  
 Memory Increment: **Enable \***  
 Peripheral Data Width: **Word \***  
 Memory Data Width: **Word \***

### LPUART1\_TX: DMA1\_Channel4 DMA request Settings:

Mode: Normal  
 Peripheral Increment: Disable  
 Memory Increment: **Enable \***  
 Peripheral Data Width: Byte  
 Memory Data Width: Byte

### ADC1: DMA1\_Channel5 DMA request Settings:

Mode: **Circular \***  
 Peripheral Increment: Disable

Memory Increment: **Enable \***  
Peripheral Data Width: **Word \***  
Memory Data Width: **Word \***

*MENTOMEM: DMA1\_Channel6 DMA request Settings:*

Mode: Normal  
Src Memory Increment: **Enable \***  
Dst Memory Increment: **Enable \***  
Src Memory Data Width: **Word \***  
Dst Memory Data Width: **Word \***

*DAC1\_CH1: DMA1\_Channel1 DMA request Settings:*

Mode: **Circular \***  
Peripheral Increment: Disable  
Memory Increment: **Enable \***  
Peripheral Data Width: Word  
Memory Data Width: **Word \***

### 4.3. NVIC configuration

#### 4.3.1. NVIC

| Interrupt Table                                                        | Enable | Preenmption Priority | SubPriority |
|------------------------------------------------------------------------|--------|----------------------|-------------|
| Non maskable interrupt                                                 | true   | 0                    | 0           |
| Hard fault interrupt                                                   | true   | 0                    | 0           |
| Memory management fault                                                | true   | 0                    | 0           |
| Prefetch fault, memory access fault                                    | true   | 0                    | 0           |
| Undefined instruction or illegal state                                 | true   | 0                    | 0           |
| System service call via SWI instruction                                | true   | 0                    | 0           |
| Debug monitor                                                          | true   | 0                    | 0           |
| Pendable request for system service                                    | true   | 15                   | 0           |
| System tick timer                                                      | true   | 15                   | 0           |
| DMA1 channel1 global interrupt                                         | true   | 5                    | 0           |
| DMA1 channel2 global interrupt                                         | true   | 5                    | 0           |
| DMA1 channel3 global interrupt                                         | true   | 5                    | 0           |
| DMA1 channel4 global interrupt                                         | true   | 5                    | 0           |
| DMA1 channel5 global interrupt                                         | true   | 5                    | 0           |
| TIM1 capture compare interrupt                                         | true   | 5                    | 0           |
| TIM7 global interrupt, DAC2 and DAC4 channel underrun error interrupts | true   | 15                   | 0           |
| COMP4, COMP5 and COMP6 interrupts through EXTI lines 30, 31 and 32     | true   | 5                    | 0           |
| LPUART1 global interrupt                                               | true   | 5                    | 0           |
| PVD/PVM1/PVM2/PVM3/PVM4 interrupts through EXTI lines 16/38/39/40/41   | unused |                      |             |
| Flash global interrupt                                                 | unused |                      |             |
| RCC global interrupt                                                   | unused |                      |             |
| DMA1 channel6 global interrupt                                         | unused |                      |             |
| ADC1 and ADC2 global interrupt                                         | unused |                      |             |
| TIM1 break interrupt and TIM15 global interrupt                        | unused |                      |             |
| TIM1 update interrupt and TIM16 global interrupt                       | unused |                      |             |
| TIM1 trigger and commutation interrupts and TIM17 global interrupt     | unused |                      |             |
| TIM2 global interrupt                                                  | unused |                      |             |
| ADC3 global interrupt                                                  | unused |                      |             |
| TIM6 global interrupt, DAC1 and DAC3 channel underrun error interrupts | unused |                      |             |
| ADC4 global interrupt                                                  | unused |                      |             |
| FPU global interrupt                                                   | unused |                      |             |

#### 4.3.2. NVIC Code generation

| Enabled interrupt Table                                                   | Select for init<br>sequence ordering | Generate IRQ<br>handler | Call HAL handler |
|---------------------------------------------------------------------------|--------------------------------------|-------------------------|------------------|
| Non maskable interrupt                                                    | false                                | true                    | false            |
| Hard fault interrupt                                                      | false                                | true                    | false            |
| Memory management fault                                                   | false                                | true                    | false            |
| Prefetch fault, memory access fault                                       | false                                | true                    | false            |
| Undefined instruction or illegal state                                    | false                                | true                    | false            |
| System service call via SWI instruction                                   | false                                | false                   | false            |
| Debug monitor                                                             | false                                | true                    | false            |
| Pendable request for system service                                       | false                                | false                   | false            |
| System tick timer                                                         | false                                | false                   | true             |
| DMA1 channel1 global interrupt                                            | false                                | true                    | true             |
| DMA1 channel2 global interrupt                                            | false                                | true                    | true             |
| DMA1 channel3 global interrupt                                            | false                                | true                    | true             |
| DMA1 channel4 global interrupt                                            | false                                | true                    | true             |
| DMA1 channel5 global interrupt                                            | false                                | true                    | true             |
| TIM1 capture compare interrupt                                            | false                                | true                    | true             |
| TIM7 global interrupt, DAC2 and DAC4<br>channel underrun error interrupts | false                                | true                    | true             |
| COMP4, COMP5 and COMP6 interrupts<br>through EXTI lines 30, 31 and 32     | false                                | true                    | true             |
| LPUART1 global interrupt                                                  | false                                | true                    | true             |

\* User modified value



## 5. System Views

### 5.1. Category view

#### 5.1.1. Current

| Middleware  |        |        |        |              |            |          |           |           |     |
|-------------|--------|--------|--------|--------------|------------|----------|-----------|-----------|-----|
| FREERTOS    |        |        |        |              |            |          |           |           |     |
| System Core | Analog |        | Timers | Connectivity | Multimedia | Security | Computing | Utilities | Bsp |
| DMA         | ADC1   | ADC2   | TIM1   | LPUART1      |            |          |           |           |     |
| GPIO        | ADC3   | ADC4   | TIM2   |              |            |          |           |           |     |
| NVIC        | COMP5  | COMP6  | TIM15  |              |            |          |           |           |     |
| RCC         | DAC1   | DAC3   |        |              |            |          |           |           |     |
| SYS         | DAC4   | OPAMP2 |        |              |            |          |           |           |     |
|             | OPAMP3 | OPAMP4 |        |              |            |          |           |           |     |
|             | OPAMP5 | OPAMP6 |        |              |            |          |           |           |     |

## 6. Docs & Resources

| Type                    | Link                                                                                                                                                                                                                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BSDL files              | <a href="https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip">https://www.st.com/resource/en/bsdl_model/stm32g4_bsd1.zip</a>                                                                                                                                                         |
| IBIS models             | <a href="https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip">https://www.st.com/resource/en/ibis_model/stm32g4_ibis.zip</a>                                                                                                                                                         |
| System View Description | <a href="https://www.st.com/resource/en/svd/stm32g4_svd.zip">https://www.st.com/resource/en/svd/stm32g4_svd.zip</a>                                                                                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf</a>                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf">https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf</a>                                                                                                         |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf">https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf</a>                                                                                   |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf">https://www.st.com/resource/en/product_presentation/stm32-usb-c-pd-solutions-presentation.pdf</a>                                                                                   |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf">https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf</a>                                                                                   |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf</a>                                                                                 |
| Presentations           | <a href="https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf">https://www.st.com/resource/en/product_presentation/microcontrollers-stm32g4-series-product-overview.pdf</a>                                                             |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf">https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf</a>                                                                                         |
| Brochures               | <a href="https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf">https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf</a>                                                                                       |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32g4.pdf">https://www.st.com/resource/en/flyer/flstm32g4.pdf</a>                                                                                                                                                                         |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32nucleo.pdf">https://www.st.com/resource/en/flyer/flstm32nucleo.pdf</a>                                                                                                                                                                 |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flstm32trust.pdf">https://www.st.com/resource/en/flyer/flstm32trust.pdf</a>                                                                                                                                                                   |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/flpowerstbd.pdf">https://www.st.com/resource/en/flyer/flpowerstbd.pdf</a>                                                                                                                                                                     |
| Flyers                  | <a href="https://www.st.com/resource/en/flyer/fldpstpf11120.pdf">https://www.st.com/resource/en/flyer/fldpstpf11120.pdf</a>                                                                                                                                                                 |
| Security Bulletin       | <a href="https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-">https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-</a> |

stmicroelectronics.pdf

Application Notes [https://www.st.com/resource/en/application\\_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application\\_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf)

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| Application Notes for related Tools | <a href="https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf</a>                                                                                                   |
| & Software                          |                                                                                                                                                                                                                                                                                                                                                                                                     |
| Application Notes for related Tools | <a href="https://www.st.com/resource/en/application_note/an5496-guidelines-for-the-buck-voltage-mode-on-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5496-guidelines-for-the-buck-voltage-mode-on-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf</a>                                                                               |
| & Software                          |                                                                                                                                                                                                                                                                                                                                                                                                     |



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| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an5497-introduction-to-the-buck-current-mode-with-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an5497-introduction-to-the-buck-current-mode-with-the-bg474edpow1-discovery-kit-stmicroelectronics.pdf</a>             |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6179-how-to-integrate-the-stl-firmware-into-a-time-critical-user-application-stmicroelectronics.pdf</a>               |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6127-getting-started-with-stm32h7rx7sx-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                                 |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6265-getting-started-with-stm32n6-mcus-in-stm32cubeide-stmicroelectronics.pdf</a>                                                           |
| Application Notes<br>for related Tools<br>& Software | <a href="https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf">https://www.st.com/resource/en/application_note/an6382-introduction-to-swd-multidrop-for-stm32-with-stlink-stmicroelectronics.pdf</a>                                                       |
| Errata Sheets                                        | <a href="https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf">https://www.st.com/resource/en/errata_sheet/es0430-stm32g471xx473xx474xx483xx484xx-device-errata-stmicroelectronics.pdf</a>                                                                           |
| Datasheet                                            | <a href="https://www.st.com/resource/en/datasheet/dm00431551.pdf">https://www.st.com/resource/en/datasheet/dm00431551.pdf</a>                                                                                                                                                                                                           |
| Programming<br>Manuals                               | <a href="https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf">https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf</a>                                                           |
| Reference<br>Manuals                                 | <a href="https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf">https://www.st.com/resource/en/reference_manual/rm0440-stm32g4-series-advanced-armbased-32bit-mcus-stmicroelectronics.pdf</a>                                                                       |
| Technical Notes<br>& Articles                        | <a href="https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1163-description-of-wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf</a>               |
| Technical Notes<br>& Articles                        | <a href="https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf</a>                   |
| Technical Notes<br>& Articles                        | <a href="https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles                        | <a href="https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf</a> |

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| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf</a>                         |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1208-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-tssop-and-ssop-packages-stmicroelectronics.pdf</a> |
| Technical Notes<br>& Articles | <a href="https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf">https://www.st.com/resource/en/technical_note/tn1433-reference-device-marking-schematics-for-stm32-microcontrollers-and-microprocessors-stmicroelectronics.pdf</a>                   |
| User Manuals                  | <a href="https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf">https://www.st.com/resource/en/user_manual/um3167-stm32g4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf</a>                                                             |